

An Analysis of Medical image computing

Medical image computing

Medical image computing -- Part 1

The transformation model. Common choices are rigid, affine, and deformable transformation models. B-spline and thin plate spline models are commonly used for parameterized transformation fields. Non-parametric or dense deformation fields carry a displacement vector at every grid location; this necessitates additional regularization constraints. A specific class of deformation fields are diffeomorphisms, which are invertible transformations with a smooth inverse. The similarity metric. A distance or similarity function is used to quantify the registration quality. This similarity can be calculated either on the original images or on features extracted from the images. Common similarity measures are sum of squared distances (SSD), correlation coefficient, and mutual information. The choice of similarity measure depends on whether the images are from the same modality; the acquisition noise can also play a role in this decision. For example, SSD is the optimal similarity measure for images of the same modality with Gaussian noise. However, the image statistics in ultrasound are significantly different from Gaussian noise, leading to the introduction of ultrasound specific similarity measures. Multi-modal registration requires a more sophisticated similarity measure; alternatively, a different image representation can be used, such as structural representations or registering adjacent anatomy. A 2020 study employed contrastive coding to learn shared, dense image representations, referred to as contrastive multi-modal image representations (CoMIRs), which enabled the registration of multi-modal images where existing registration methods often fail due to a lack of sufficiently similar image structures. It reduced the multi-modal registration problem to a mono-modal one, in which general intensity based, as well as feature-based, registration algorithms can be applied. The optimization procedure. Either continuous or discrete optimization is performed. For continuous optimization, gradient-based optimization techniques are applied to improve the convergence speed.

Medical image computing -- Part 2

Modality-specific computing Some imaging modalities provide very specialized information. The resulting images cannot be treated as regular scalar images and give rise to new sub-areas of medical image computing. Examples include diffusion MRI and functional MRI.

Medical image computing -- Part 3

Software Software for medical image computing is a complex combination of systems providing IO, visualization and interaction, user interface, data management and computation. Typically system architectures are layered to serve algorithm

developers, application developers, and users. The bottom layers are often libraries and/or toolkits which provide base computational capabilities; while the top layers are specialized applications which address specific medical problems, diseases, or body systems.

Medical image computing -- Part 4

Statistical analysis Statistical methods combine the medical imaging field with modern computer vision, machine learning and pattern recognition. Over the last decade, several large datasets have been made publicly available (see for example ADNI, 1000 functional Connectomes Project), in part due to collaboration between various institutes and research centers. This increase in data size calls for new algorithms that can mine and detect subtle changes in the images to address clinical questions. Such clinical questions are very diverse and include group analysis, imaging biomarkers, disease phenotyping and longitudinal studies.

Medical image computing -- Part 5

Multiple templates Rather than relying on a single template, multiple templates can be used. The idea is to represent an image as a deformed version of one of the templates. For example, there could be one template for a healthy population and one template for a diseased population. However, in many applications, it is not clear how many templates are needed. A simple albeit computationally expensive way to deal with this is to have every image in a training dataset be a template image and thus every new image encountered is compared against every image in the training dataset. A more recent approach automatically finds the number of templates needed.

Medical image computing -- Part 6

Image registration is a process that searches for the correct alignment of images. In the simplest case, two images are aligned. Typically, one image is treated as the target image and the other is treated as a source image; the source image is transformed to match the target image. The optimization procedure updates the transformation of the source image based on a similarity value that evaluates the current quality of the alignment. This iterative procedure is repeated until a (local) optimum is found. An example is the registration of CT and PET images to combine structural and metabolic information (see figure). Image registration is used in a variety of medical applications:

Medical image computing -- Part 7

An Analysis of Medical image computing

Medical image computing

Segmentation is the process of partitioning an image into different meaningful segments. In medical imaging, these segments often correspond to different tissue classes, organs, pathologies, or other biologically relevant structures. Medical image segmentation is made difficult by low contrast, noise, and other imaging ambiguities. Although there are many computer vision techniques for image segmentation, some have been adapted specifically for medical image computing. Below is a sampling of techniques within this field; the implementation relies on the expertise that clinicians can provide.

Medical image computing -- Part 8

Additional notes Medical image computing is also related to the field of computer vision. An international society, The MICCAI Society, represents the field and organizes an annual conference and associated workshops. Proceedings for this conference are published by Springer in the Lecture Notes in Computer Science series. In 2000, N. Ayache and J. Duncan reviewed the state of the field.

Medical image computing -- Part 9

Radiology the official journal of the Radiological Society of North America NeuroImage Journal of Magnetic Resonance Imaging (JMRI) Magnetic Resonance in Medicine (MRM) Journal of Computer Assisted Tomography (JCAT) Human Brain Mapping

Medical image computing -- Part 10

Atlas-based segmentation: For many applications, a clinical expert can manually label several images; segmenting unseen images is a matter of extrapolating from these manually labeled training images. Methods of this style are typically referred to as atlas-based segmentation methods. Parametric atlas methods typically combine these training images into a single atlas image, while nonparametric atlas methods typically use all of the training images separately. Atlas-based methods usually require the use of image registration in order to align the atlas image or images to a new, unseen image. Shape-based segmentation: Many methods parametrize a template shape for a given structure, often relying on control points along the boundary. The entire shape is then deformed to match a new image. Two of the most common shape-based techniques are active shape models and active appearance models. These methods have been very influential, and have given rise to similar models. Image-based segmentation: Some methods initiate a template and refine its shape according to the image data while minimizing integral error measures, like the active contour model and its variations. Interactive segmentation: Interactive methods are useful when

clinicians can provide some information, such as a seed region or rough outline of the region to segment. An algorithm can then iteratively refine such a segmentation, with or without guidance from the clinician. Manual segmentation, using tools such as a paint brush to explicitly define the tissue class of each pixel, remains the gold standard for many imaging applications. Recently, principles from feedback control theory have been incorporated into segmentation, which give the user much greater flexibility and allow for the automatic correction of errors. Subjective surface segmentation: This method is based on the idea of evolution of segmentation function which is governed by an advection-diffusion model. To segment an object, a segmentation seed is needed (that is the starting point that determines the approximate position of the object in the image). Consequently, an initial segmentation function is constructed. The idea behind the subjective surface method is that the position of the seed is the main factor determining the form of this segmentation function. Convolutional neural networks (CNNs): The computer-assisted fully automated segmentation performance has been improved due to the advancement of machine learning models. CNN based models such as SegNet, UNet, ResNet, AATSN, Transformers and GANs have fastened the segmentation process. In the future, such models may replace manual segmentation due to their superior performance and speed. There are other classifications of image segmentation methods that are similar to categories above. Another group, which is based on combination of methods, can be classified as "hybrid".